Yihan Li

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Education

Sun Yat-sen University Sep 2023 – Jul 2027 (Expected)
B.A. in English Language & Literature, School of Foreign Languages *Major GPA: 3.8/4.0*
Minor in Intelligent Unmanned Systems *GPA: 4.0/4.0*
Minor in Computer Science & Technology *GPA: 3.8/4.0*
Relevant Coursework: Fundamentals of Computer Technology (95), Mathematical Modeling and Practice (92), MATLAB Computation and Simulation (90), Data Structures and Algorithms Laboratory (89), Natural Language Processing (86).

Publications

Anticipating Contact for Visuo-Tactile Diffusion Policies Under Review
Yating Feng, Yuhang Zheng, Lixin Xu, **Yihan Li**, Jiaju Yin, Renjing Xu

Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation Under Review
Yihan Li, Hanyi Zhang, Xiaoxi Jiang, Man Guo

Unordered Landmark Visual Navigation ECCV 2026
Hao Ren, Junzhe Zhu, **Yihan Li**, Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng

OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models IROS 2026
Yihan Li, Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng

RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy NeurIPS 2025
Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, **Yihan Li**, Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng

Professional Experience

HCL Lab (HKUST(GZ)) Jan 2026 – Present
Research Assistant *Prof. Renjing Xu*

- Researching tactile sensing, vision models, VLA fine-tuning, VLA-based online RL, and Real2Sim scene reproduction for dexterous manipulation.
- Exploring multimodal visuo-tactile policy learning and evaluation protocols for embodied manipulation under real-world sensing constraints.

Yun Drone Feb 2025 – Present
UAV Development Intern

- Worked on UAV autonomy development across propulsion selection, formation flight, navigation and localization, and flight-test workflow design.
- Studied RL-based disturbance-resilient control and supported system iteration through flight-log analysis and control tuning.

RAPID Lab (SYSU) Nov 2024 – Mar 2026
Research Assistant *Prof. Hui Cheng*

- Conducted embodied intelligence research across dexterous manipulation, multimodal perception, and vision-based navigation/traversability.
- Supported RAPID Hand, OmniTrav, and ULVN research directions through experimental validation, perception integration, and benchmark development.

Leadership Experience

RoboTech Association (SYSU) Sep 2024 – Dec 2025

President; Competition Team Captain

- Led robotics training and competition preparation across mechanical design, hardware, embedded control, vision, navigation, and algorithms.
- Coordinated association operations, technical mentoring, cross-group collaboration, and onboarding materials for 160+ members.

Projects

Venom VNV: ROS 2 Unified Robotics Platform Nov 2025 – Present

Project Lead; ROS 2 Platform Developer venom-algorithm.github.io/Venom_VNV

- Architected a ROS 2 Humble platform spanning drivers, perception, localization, planning, mission control, simulation, and interface specifications for UGV/UAV/USV scenarios.
- Integrated Livox MID360, Hikvision cameras, AgileX chassis/arm/PX4 drivers with Point-LIO/Fast-LIO, Nav2 TEB, Ego Planner Swarm, YOLO detection, auto-aim tracking, and waypoint task management.

RAPID Hand: Perception-Integrated Dexterous Manipulation Platform Nov 2024 – Jun 2025

Research Assistant rapid-hand.github.io

- Co-developed a compact 20-DoF dexterous hand ontology and high-DoF teleoperation/data-collection stack for contact-rich whole-hand manipulation.
- Integrated hardware-level whole-hand perception, synchronizing wrist vision, fingertip tactile sensing, and proprioception with sub-7 ms latency for policy learning.

OmniTrav: 360° **Vision-based Traversability Benchmark** Jun 2025 – Mar 2026

First Author

- Proposed the C2T pipeline and built OmniTrav, a 4.7k-scene, 250k-frame, 577 GB dataset mapping pinhole, fisheye, and panoramic RGB to planner-ready 360-degree radial clearance.
- Designed an automated foundation-model annotation pipeline and a DINOv3-FPN baseline with a cross-attention angle projector for radial distance and obstacle prediction.

GenreShift and Rosetta: LLM Systems for Linguistic Research Oct 2024 – Present

Project Leader genreshift.cn
rosetta-stone.xyz

- Built GenreShift for genre and discipline analysis, extracting quantitative linguistic features such as TTR, MTLD, academic vocabulary, and cross-genre comparisons for LoRA/MoE-based genre transformation; filed invention patent application No. 2025105553019.
- Developed Rosetta as an agentic annotation pipeline with concept bootstrap, prompt optimization, embedding-based top-k retrieval, LLM judge routing, review feedback, and Prodigy-compatible JSONL export.

Awards

China Robot and Artificial Intelligence Competition 2025

National First Prize (Team Captain)

National Intelligent Unmanned Systems Application Challenge 2025

National Champion (Team Captain)

National Undergraduate Electronics Design Contest 2025

National Second Prize (Team Captain)

National Undergraduate Robotics Competition - RoboMaster University League 2025

National Second Prize (Vision Group Leader)

Skills

Programming:	Python, C/C++, MATLAB, STM32/Arduino.
Robotics and Sim:	ROS1/2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, PX4, Hermes, OpenClaw.
Development Tools:	Codex, Claude Code.
Engineering Tools:	Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot.
Languages:	Mandarin (native), English (TEM-4), French.